

10 Mathematical Induction

Prove the following statements with either induction, strong induction or proof by smallest counterexample.

2. For every integer $n \in \mathbb{N}$, it follows that $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1^2 = \frac{(1)(2)(3)}{6} = 1$, which is true.
2. Assume $1^2 + 2^2 + 3^2 + 4^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$ for some integer $k \geq 1$.
 Observe that this implies the statement is true for $n = k + 1$.

$$\begin{aligned}
 & 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 \\
 &= (1^2 + 2^2 + 3^2 + \dots + k^2) + (k+1)^2 \\
 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\
 &= \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6} \\
 &= \frac{k(k+1)(2k+1) + 6(k+1)(k+1)}{6} \\
 &= \frac{2k^3 + 9k^2 + 13k + 6}{6} \\
 &= \frac{(k+1)(k+2)(2k+3)}{6} \\
 &= \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}
 \end{aligned}$$

Therefore $1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}$,
which means the statement is true for $n = k+1$.

It follows by induction that the statement is true for every $n \in \mathbb{N}$. $\square$

4. If $n \in \mathbb{N}$, then $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1(1+1) = \frac{1(1+1)(1+2)}{3} = 2$, which is true.
2. Assume $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$ for some integer $k \geq 1$.
Observe that this implies the statement is true for $n = k+1$.

$$\begin{aligned}
& 1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2) \\
&= (1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1)) + (k+1)(k+2) \\
&= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\
&= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3} \\
&= \frac{k(k^2 + 3k + 2) + 3(k^2 + 3k + 2)}{3} \\
&= \frac{k^3 + 6k^2 + 11k + 6}{3} \\
&= \frac{(k+1)(k^2 + 5k + 6)}{3} \\
&= \frac{(k+1)(k+2)(k+3)}{3} \\
&= \frac{(k+1)([k+1]+1)([k+1]+2)}{3}
\end{aligned}$$

Therefore $1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2) = \frac{(k+1)([k+1]+1)([k+1]+2)}{3}$,
which means the statement is true for $n = k+1$.

It follows by induction that the statement is true for every $n \in \mathbb{N}$. $\square$

6. For every natural number n , it follows that $\sum_{i=1}^n (8i - 5) = 4n^2 - n$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, then the statement is $\sum_{i=1}^1 (8i - 5) = 4(1^2) - 1 = 3$, which is true.
2. Assume $\sum_{i=1}^k (8i - 5) = 4k^2 - k$ for some $k \geq 1$. Then

$$\begin{aligned}
\sum_{i=1}^{k+1} (8i - 5) &= \left[\sum_{i=1}^k (8i - 5) \right] + \left[8(k+1) - 5 \right] \\
&= [4k^2 - k] + [8(k+1) - 5] \\
&= 4k^2 - k + 8k + 3 \\
&= 4k^2 + 8k + 4 - k - 1 \\
&= 4(k^2 + 2k + 1) - (k + 1) \\
&= 4(k+1)(k+1) - (k+1) \\
&= 4(k+1)^2 - (k+1)
\end{aligned}$$

This means the statement is true for $n = k + 1$.

It follows by induction that $\sum_{i=1}^n (8i - 5) = 4n^2 - n$ for every natural number n . □

8. If $n \in \mathbb{N}$, then $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, then the statement is $\frac{1}{2!} = 1 - \frac{1}{2!} = \frac{1}{2}$, which is true.

2. Assume $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$ for some $k \geq 1$. Then

$$\begin{aligned}
& \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} \\
&= \left[\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} \right] + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{k+2}{(k+2)(k+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{k+2}{(k+2)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{(k+2) - (k+1)}{(k+2)!} \\
&= 1 - \frac{1}{(k+2)!} \\
&= 1 - \frac{1}{([k+1] + 1)!}
\end{aligned}$$

This means the statement is true for $n = k + 1$.

It follows by induction that $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$ for every $n \in \mathbb{N}$. $\square$

10. For any integer $n \geq 0$, it follows that $3 \mid (5^{2n} - 1)$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 0$, then the statement is $3 \mid 0$, which is true.
2. Assume this statement is true for some integer $n = k \geq 0$. That is, assume $3 \mid (5^{2k} - 1)$. This means $5^{2k} - 1 = 3a$ for some integer a , and from this we get $5^{2k} = 3a + 1$. Now observe that

$$\begin{aligned}
5^{2(k+1)} - 1 &= 5^{2k+2} - 1 \\
&= 5^2 \cdot 5^{2k} - 1 \\
&= 5^2(3a + 1) - 1 \\
&= 25(3a + 1) - 1 \\
&= 25 \cdot 3a + 25 - 1 \\
&= 3(25a + 8)
\end{aligned}$$

This shows $5^{2(k+1)} - 1 = 3(25a + 8)$. Since $25a + 8 \in \mathbb{Z}$, it follows that $3 \mid (5^{2(k+1)} - 1)$.

It follows by induction that $3 \mid (5^{2n} - 1)$ for every integer $n \geq 0$. $\square$

12. For any integer $n \geq 0$, it follows that $9 \mid (4^{3n} + 8)$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 0$, then the statement is $9 \mid 9$, which is true.
2. Assume this statement is true for some integer $n = k \geq 0$. That is, assume $9 \mid (4^{3k} + 8)$. This means $4^{3k} + 8 = 9a$ for some integer a , and from this we get $4^{3k} = 9a - 8$. Now observe that

$$\begin{aligned} 4^{3(k+1)} + 8 &= 4^{3k+3} + 8 \\ &= 4^3 \cdot 4^{3k} + 8 \\ &= 4^3(9a - 8) + 8 \\ &= 64(9a - 8) + 8 \\ &= 64(9a) - 512 + 8 \\ &= 9(64a - 56) \end{aligned}$$

This shows $4^{3(k+1)} + 8 = 9(64a - 56)$. Since $64a - 56 \in \mathbb{Z}$, it follows that $9 \mid (4^{3(k+1)} + 8)$.

It follows by induction that $9 \mid (4^{3n} + 8)$ for every integer $n \geq 0$. □

14. Suppose $a \in \mathbb{Z}$. Prove that $5 \mid 2^n a$ implies $5 \mid a$ for any $n \in \mathbb{N}$.

Proof. The proof is induction on n .

1. Let $n = 1$. Suppose $5 \mid 2^1 a$. Since 5 is prime, we have $5 \mid 2$ or $5 \mid a$. But $5 \nmid 2$, therefore $5 \mid a$.
2. Assume this statement is true for some integer $n = k \geq 1$. That is, assume $5 \mid 2^k a$ implies $5 \mid a$. We must show that $5 \mid 2^{k+1} a$ implies $5 \mid a$. Suppose $5 \mid 2^{k+1} a$. We can write $2^{k+1} a = 2^k(2a)$. Therefore $5 \mid 2^k(2a)$. By the induction hypothesis, we have $5 \mid 2a$. By the base case, we have $5 \mid a$. Therefore, the statement holds for $k + 1$.

It follows by induction that $5 \mid 2^n a$ implies $5 \mid a$ for every $n \in \mathbb{N}$. □

16. For every natural number n , it follows that $2^n + 1 \leq 3^n$.

Proof. We prove the statement by induction.

1. Observe that if $n = 1$, then the statement is $2^1 + 1 \leq 3^1$, that is, $3 \leq 3$, which is true.

2. Assume this statement is true for some natural number $k \geq 1$. That is, assume $2^k + 1 \leq 3^k$. We need to show that $2^{k+1} + 1 \leq 3^{k+1}$. By the induction hypothesis, $2^k \leq 3^k - 1$. Observe that

$$\begin{aligned} 2^{k+1} + 1 &= 2(2^k) + 1 \leq 2(3^k - 1) + 1 = 2(3^k) - 1 && \text{(induction hypothesis)} \\ &\leq 3(3^k) = 3^{k+1} && \text{(since } -1 < 3^k \text{)} \end{aligned}$$

This shows the statement is true for $n = k + 1$.

It follows by induction that $2^n + 1 \leq 3^n$ for every natural number n . $\square$

18. Suppose $A_1, A_2, \dots, A_n$ are sets in some universal set U , and $n \geq 2$. Prove that $\overline{A_1 \cup A_2 \cup \dots \cup A_n} = \overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}$.

Proof. The proof is by strong induction.

- When $n = 2$ the statement is $\overline{A_1 \cup A_2} = \overline{A_1} \cap \overline{A_2}$, which is true (see Chapter 8, Exercise 11).
- Let $k \geq 2$. Assume the statement is true if it involves k or fewer sets. Then

$$\begin{aligned} &\overline{A_1 \cup A_2 \cup \dots \cup A_{k-1} \cup A_k \cup A_{k+1}} \\ &= \overline{A_1 \cup A_2 \cup \dots \cup A_{k-1} \cup (A_k \cup A_{k+1})} && \text{(regrouping)} \\ &= \overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_{k-1}} \cap \overline{A_k \cup A_{k+1}} && \text{(inductive hypothesis, } k \text{ sets)} \\ &= \overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_{k-1}} \cap \overline{A_k} \cap \overline{A_{k+1}} && \text{(two-set case)} \end{aligned}$$

Thus the statement is true when it involves $k + 1$ sets.

This completes the proof by strong induction. $\square$

20. Prove that $(1 + 2 + 3 + \dots + n)^2 = 1^3 + 2^3 + 3^3 + \dots + n^3$ for every $n \in \mathbb{N}$.

Proof. We prove the statement by induction.

- Observe that if $n = 1$, then the statement is $1^2 = 1^3$, that is, $1 = 1$, which is true.
- Assume $(1 + 2 + 3 + \dots + k)^2 = 1^3 + 2^3 + 3^3 + \dots + k^3$ for some natural number $k \geq 1$. We need to show that $(1 + 2 + 3 + \dots + k + (k + 1))^2 = 1^3 + 2^3 + 3^3 + \dots + k^3 + (k + 1)^3$.

Let $T = 1 + 2 + 3 + \dots + k$; then $T = \frac{k(k+1)}{2}$ (see Exercise 1), so $2T = k(k+1)$. Observe that

$$\begin{aligned}
& (1 + 2 + 3 + \dots + k + (k+1))^2 \\
&= (T + (k+1))^2 \\
&= T^2 + 2T(k+1) + (k+1)^2 \\
&= 1^3 + 2^3 + \dots + k^3 + 2T(k+1) + (k+1)^2 \quad (\text{induction hypothesis}) \\
&= 1^3 + 2^3 + \dots + k^3 + k(k+1)(k+1) + (k+1)^2 \quad (\text{since } 2T = k(k+1)) \\
&= 1^3 + 2^3 + \dots + k^3 + (k+1)^2(k+1) \\
&= 1^3 + 2^3 + \dots + k^3 + (k+1)^3
\end{aligned}$$

This shows the statement is true for $n = k + 1$.

It follows by induction that $(1 + 2 + 3 + \dots + n)^2 = 1^3 + 2^3 + 3^3 + \dots + n^3$ for every $n \in \mathbb{N}$. $\square$

22. If $n \in \mathbb{N}$, then $(1 - \frac{1}{2})(1 - \frac{1}{4})(1 - \frac{1}{8})(1 - \frac{1}{16}) \dots (1 - \frac{1}{2^n}) \geq \frac{1}{4} + \frac{1}{2^{n+1}}$.

Proof. We prove the statement by induction.

1. Observe that if $n = 1$, then the statement is $1 - \frac{1}{2} \geq \frac{1}{4} + \frac{1}{2^2}$, that is, $\frac{1}{2} \geq \frac{1}{2}$, which is true.
2. Assume $(1 - \frac{1}{2})(1 - \frac{1}{4}) \dots (1 - \frac{1}{2^k}) \geq \frac{1}{4} + \frac{1}{2^{k+1}}$ for some natural number $k \geq 1$. We need to show that $(1 - \frac{1}{2})(1 - \frac{1}{4}) \dots (1 - \frac{1}{2^{k+1}}) \geq \frac{1}{4} + \frac{1}{2^{k+2}}$. Let $x = \frac{1}{2^{k+1}}$. Since $k \geq 1$, we have $2^{k+1} \geq 4$, hence $0 < x \leq \frac{1}{4}$; in particular $1 - x > 0$ and $x(\frac{1}{4} - x) \geq 0$. Observe that

$$\begin{aligned}
& (1 - \frac{1}{2})(1 - \frac{1}{4}) \dots (1 - \frac{1}{2^k})(1 - \frac{1}{2^{k+1}}) \\
& \geq (\frac{1}{4} + x)(1 - x) = \frac{1}{4} + \frac{x}{2} + x(\frac{1}{4} - x) \quad (\text{induction hypothesis, } 1 - x > 0) \\
& \geq \frac{1}{4} + \frac{x}{2} = \frac{1}{4} + \frac{1}{2^{k+2}} \quad (\text{since } x(\frac{1}{4} - x) \geq 0)
\end{aligned}$$

This shows the statement is true for $n = k + 1$.

It follows by induction that the statement is true for every $n \in \mathbb{N}$. $\square$

24. Prove that $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$ for each natural number n .

Proof. We prove the statement by induction.

1. Observe that if $n = 1$, then the statement is $\sum_{k=1}^1 k \binom{1}{k} = 1 \binom{1}{1} = 1 = 1 \cdot 2^0$, which is true.

2. Assume $\sum_{k=1}^m k \binom{m}{k} = m2^{m-1}$ for some natural number $m \geq 1$. We need to show that $\sum_{k=1}^{m+1} k \binom{m+1}{k} = (m+1)2^m$. By Pascal's rule $\binom{m+1}{k} = \binom{m}{k} + \binom{m}{k-1}$, and $\binom{m}{m+1} = 0$. Observe that

$$\begin{aligned}
\sum_{k=1}^{m+1} k \binom{m+1}{k} &= \sum_{k=1}^{m+1} k \binom{m}{k} + \sum_{k=1}^{m+1} k \binom{m}{k-1} \\
&= \sum_{k=1}^m k \binom{m}{k} + \sum_{j=0}^m (j+1) \binom{m}{j} \\
&= \sum_{k=1}^m k \binom{m}{k} + \sum_{j=1}^m j \binom{m}{j} + \sum_{j=0}^m \binom{m}{j} \\
&= m2^{m-1} + m2^{m-1} + 2^m \\
&= m2^m + 2^m \\
&= (m+1)2^m
\end{aligned}$$

This shows the statement is true for $n = m + 1$.

It follows by induction that $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$ for each natural number n . □

26. Concerning the Fibonacci sequence, prove that $\sum_{k=1}^n F_k^2 = F_n F_{n+1}$.

Proof. We prove the statement by induction.

1. Observe that if $n = 1$, then the statement is $\sum_{k=1}^1 F_k^2 = F_1 F_2$, this means $1^2 = 1 \cdot 1$, which is true.
2. Assume $\sum_{k=1}^m F_k^2 = F_m F_{m+1}$ for some natural number $m \geq 1$. We need to show that $\sum_{k=1}^{m+1} F_k^2 = F_{m+1} F_{m+2}$. Observe that

$$\begin{aligned}
\sum_{k=1}^{m+1} F_k^2 &= \sum_{k=1}^m F_k^2 + F_{m+1}^2 \\
&= F_m F_{m+1} + F_{m+1} F_{m+1} && \text{(inductive hypothesis)} \\
&= F_{m+1} (F_m + F_{m+1}) \\
&= F_{m+1} F_{m+2} && \text{(definition of Fibonacci Sequence)}
\end{aligned}$$

This shows the statement is true for $n = m + 1$.

It follows by induction that $\sum_{k=1}^n F_k^2 = F_n F_{n+1}$ for each natural number n . $\square$

28. Concerning the Fibonacci sequence, prove that $F_2 + F_4 + F_6 + F_8 + \dots + F_{2n} = F_{2n+1} - 1$.

Proof. We prove the statement by induction.

1. Observe that if $n = 1$, then the statement is $F_2 = F_3 - 1$, this means $1 = 2 - 1$, which is true.
2. Assume $F_2 + F_4 + F_6 + F_8 + \dots + F_{2k} = F_{2k+1} - 1$ for some natural number k . We need to show that $F_2 + F_4 + F_6 + F_8 + \dots + F_{2(k+1)} = F_{2(k+1)+1} - 1$. Observe that

$$\begin{aligned}
 & F_2 + F_4 + F_6 + \dots + F_{2k} + F_{2(k+1)} \\
 &= F_{2k+1} - 1 + F_{2(k+1)} && \text{(inductive hypothesis)} \\
 &= F_{2k+1} + F_{2k+2} - 1 \\
 &= F_{2k+3} - 1 && \text{(definition of Fibonacci Sequence)} \\
 &= F_{2(k+1)+1} - 1
 \end{aligned}$$

This shows the statement is true for $n = k + 1$.

It follows by induction that $F_2 + F_4 + F_6 + F_8 + \dots + F_{2n} = F_{2n+1} - 1$ for each natural number n . $\square$

30. Here F_n is the n th Fibonacci number. Prove that

$$F_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}$$

Note: This is Binet's formula and many proofs can be found online.

Proof. Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$, so the statement reads $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$. First observe that φ and ψ are precisely the two roots of $x^2 - x - 1 = 0$, therefore $\varphi^2 = \varphi + 1$ and $\psi^2 = \psi + 1$. We now prove the statement by strong induction.

1. Observe that if $n = 1$, then the statement is $F_1 = \frac{\varphi - \psi}{\sqrt{5}}$. Since $\varphi - \psi = \frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = \sqrt{5}$, the right-hand side equals $1 = F_1$, so the statement is true.

If $n = 2$, then the statement is $F_2 = \frac{\varphi^2 - \psi^2}{\sqrt{5}}$. Since $\varphi^2 - \psi^2 = (\varphi + 1) - (\psi + 1) = \varphi - \psi = \sqrt{5}$, the right-hand side equals $1 = F_2$, and the statement is true.

2. Let $k \geq 2$ and assume $F_m = \frac{\varphi^m - \psi^m}{\sqrt{5}}$ for every natural number m with $1 \leq m \leq k$.

We need to show that $F_{k+1} = \frac{\varphi^{k+1} - \psi^{k+1}}{\sqrt{5}}$. Since $k \geq 2$, both k and $k-1$ lie in the range covered by the inductive hypothesis, so

$$\begin{aligned}
 F_{k+1} &= F_k + F_{k-1} && \text{(definition of Fibonacci Sequence)} \\
 &= \frac{\varphi^k - \psi^k}{\sqrt{5}} + \frac{\varphi^{k-1} - \psi^{k-1}}{\sqrt{5}} && \text{(inductive hypothesis)} \\
 &= \frac{\varphi^{k-1}(\varphi + 1) - \psi^{k-1}(\psi + 1)}{\sqrt{5}} \\
 &= \frac{\varphi^{k-1}\varphi^2 - \psi^{k-1}\psi^2}{\sqrt{5}} \\
 &= \frac{\varphi^{k+1} - \psi^{k+1}}{\sqrt{5}}.
 \end{aligned}$$

This shows the statement is true for $n = k + 1$.

It follows by strong induction that $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}} = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}$ for each natural number n . $\square$

32. Prove that the number of n -digit binary numbers that have no consecutive 1's is the Fibonacci number F_{n+2} . For example, for $n = 2$ there are three such numbers (00, 01, and 10), and $3 = F_{2+2} = F_4$. Also, for $n = 3$ there are five such numbers (000, 001, 010, 100, 101), and $5 = F_{3+2} = F_5$.

Proof. Let A_n be the number of n -digit binary strings containing no consecutive 1's. Then $A_n = F_{n+2}$ for every natural number $n \geq 1$. We will prove that $A_n = F_{n+2}$ for all $n \geq 1$ by strong induction.

1. Observe that if $n = 1$ then the valid strings are 0 and 1, so $A_1 = F_3 = 2$ which is true.

Observe that if $n = 2$ then the valid strings are 00, 01, and 10, so $A_2 = F_4 = 3$ which is true.

Thus the statement holds for $n = 1$ and $n = 2$.

2. Assume $A_k = F_{k+2}$ for some integer k such as $1 \leq k \leq n$. We need to show that $A_{k+1} = F_{k+3}$.

Consider any valid binary string of length $k+1$, this string can begin with either a 1 or a 0. If it begins with a 0, then the remaining k digits can be any valid k -digit binary string, thus there are A_k such strings. If it begins with a 1, then the second

digit must be 0, thus the string begins with 10 and the remaining $k - 1$ digits can be any valid $(k - 1)$ -digit binary string, thus there are A_{n-1} such strings. Therefore $A_{k+1} = A_k + A_{k-1}$. By the induction hypothesis $A_k = F_{n+2}$ and $A_{k-1} = F_{k+1}$. Therefore $A_{k+1} = F_{k+2} + F_{k+1}$ and by the Fibonacci recurrence, $A_{k+1} = F_{k+3}$. This shows that statement holds for $k + 1$.

It follows by strong induction that $A_n = F_{n+2}$ for every natural number n . □